

GFAz: order-of-magnitude pangenome analytics by computing over compression

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84×
smaller
369 GB → 4.5 GB

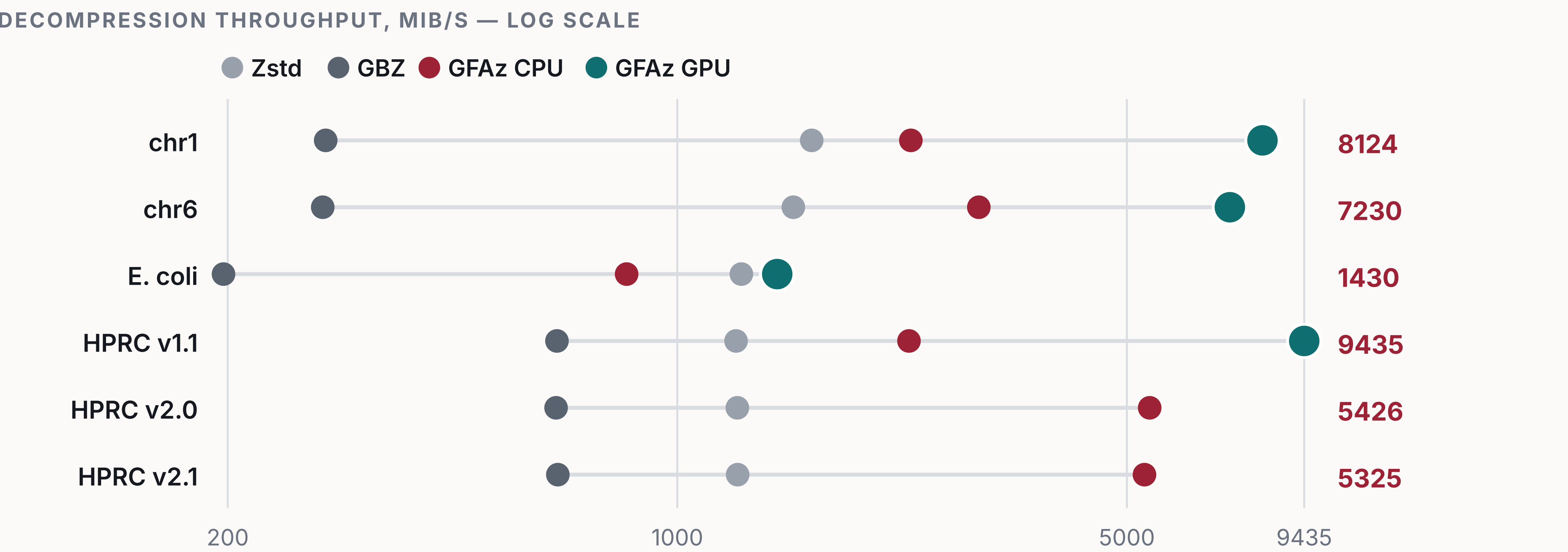
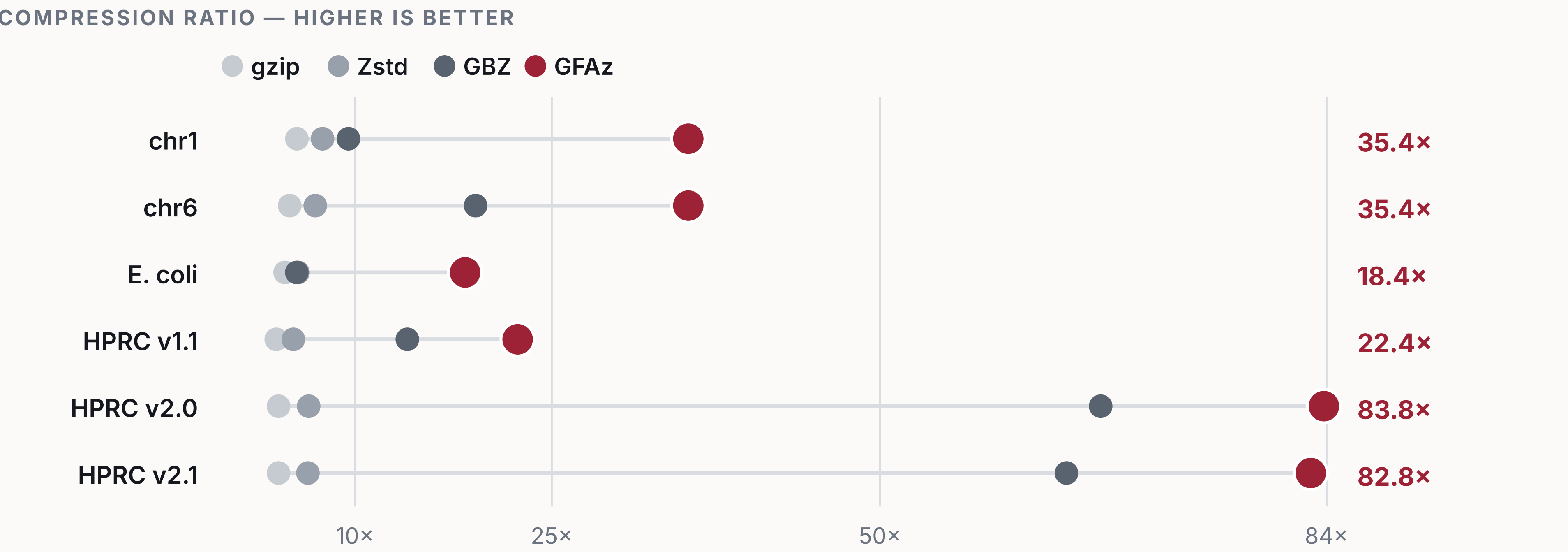
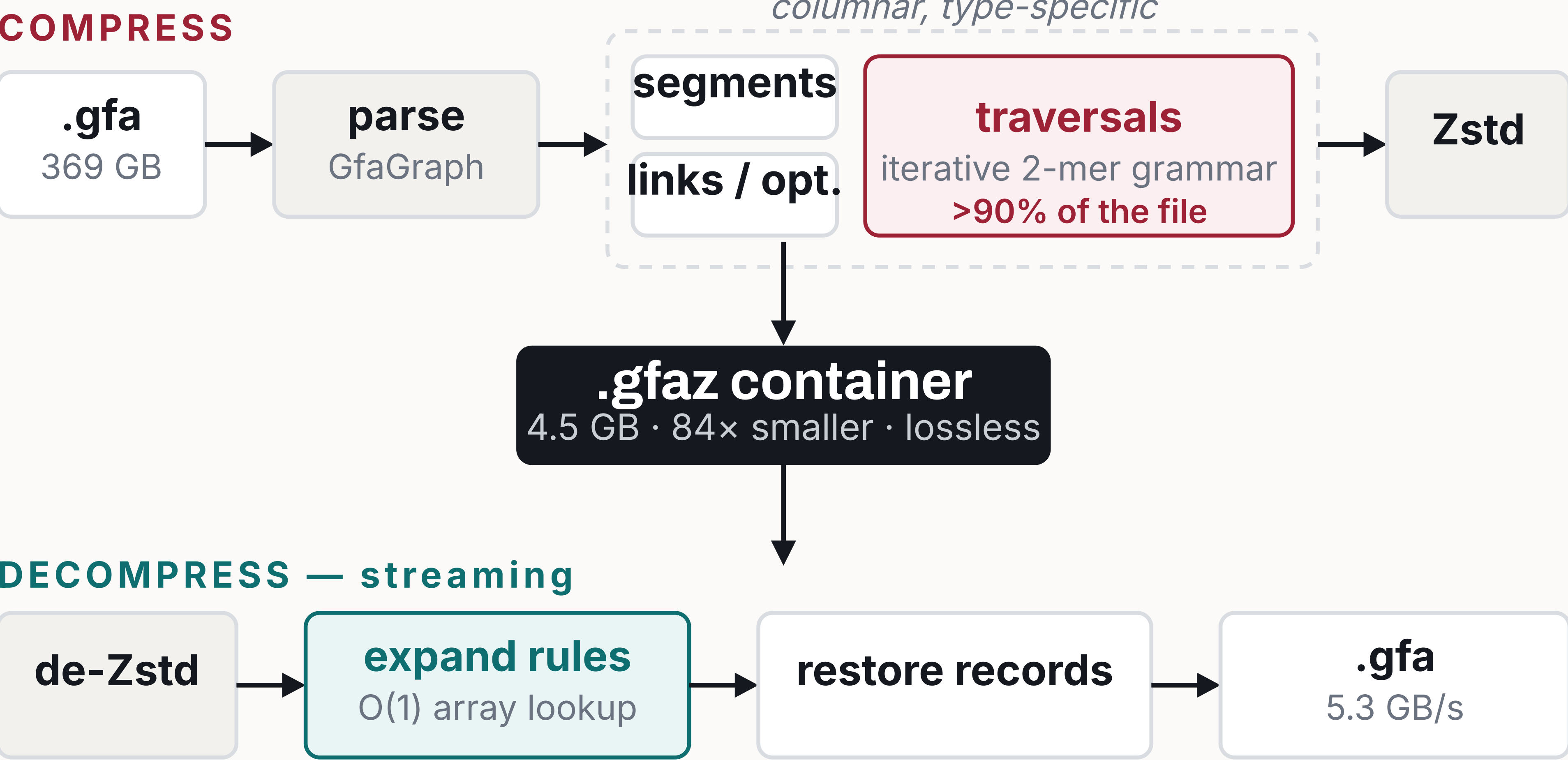
5.3 GB/s
decompression
CPU, 32 threads

369×
faster analytics
245 min → 40 s

99.99%
VCF concordance
vs. vg deconstruct

01 Compression enginelossless · full GFA semantics · CPU + GPU

02 Compute enginerun the analysis on the container

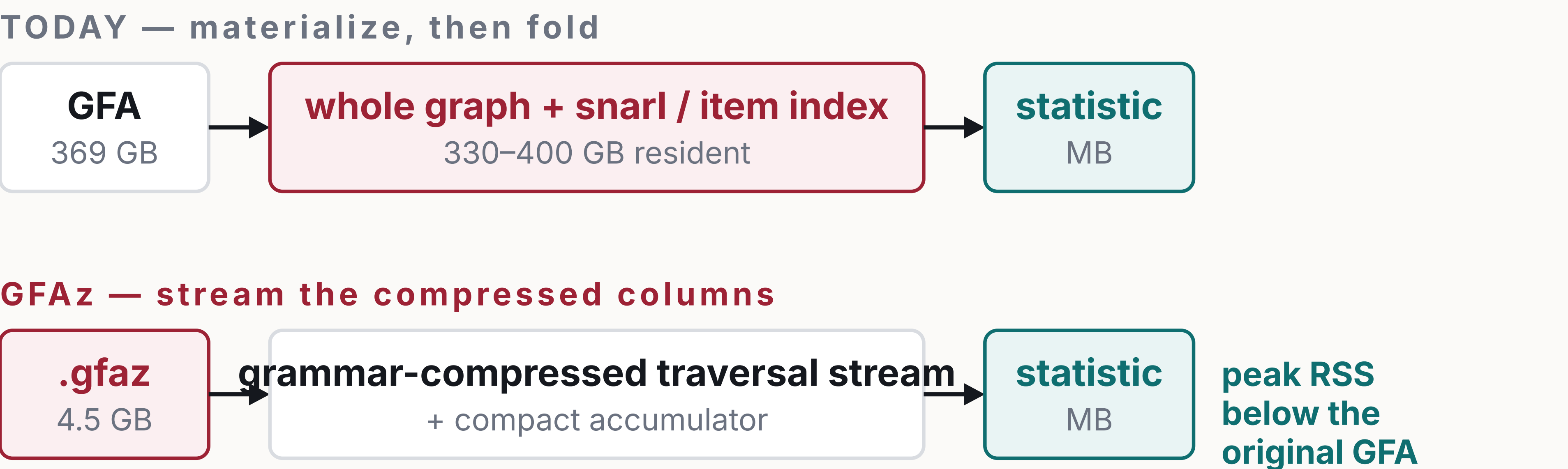


Compression reaches **1355 MiB/s** on CPU (32 threads, end-to-end CLI wall time) and **4843 MiB/s** on GPU. GFAz wins ratio on **every** dataset while decompressing faster than Zstd — and unlike SQZ and GBZ it preserves the complete GFA.

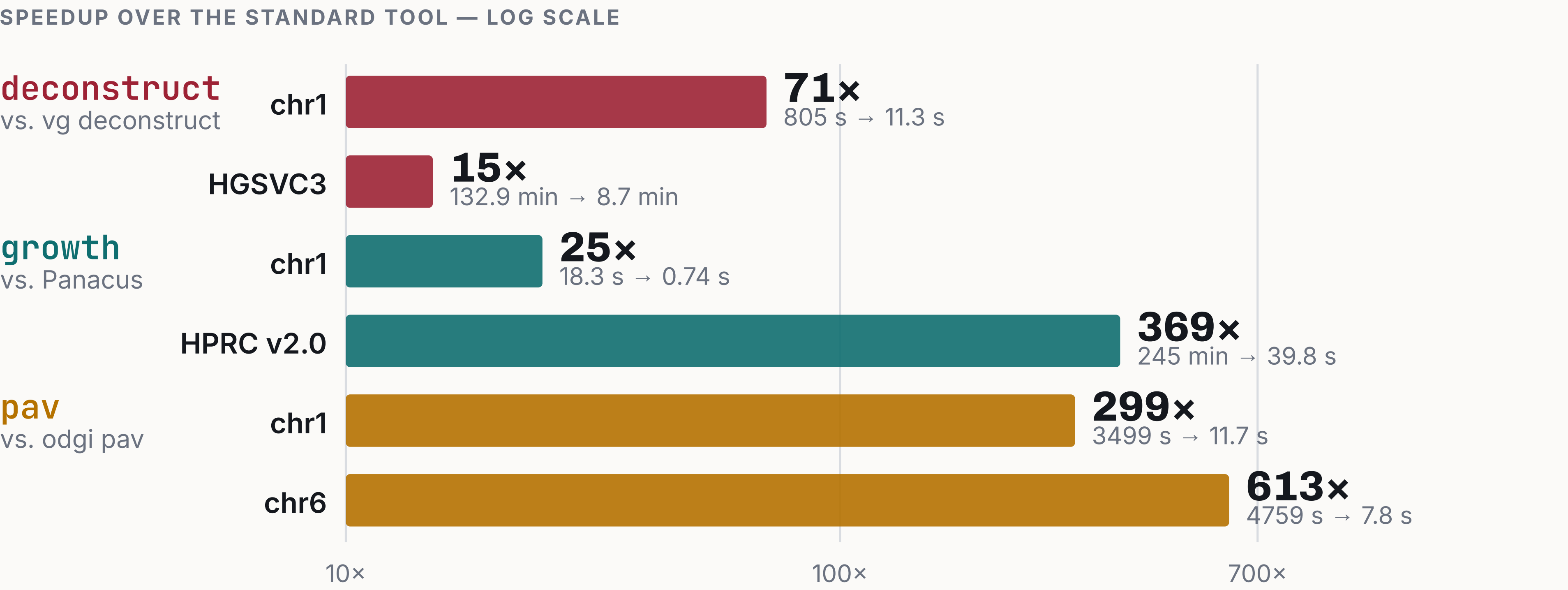
GFAZ CPU, END TO END				
GRAPH	GFA	RATIO	COMPRESS	DECOMPRESS
chr1 (PGGB)	7.1 GB	35.4×	1320 MiB/s	2307 MiB/s
HPRC v1.1	48 GB	22.4×	291 MiB/s	2292 MiB/s
HPRC v2.1	369 GB	82.8×	538 MiB/s	5325 MiB/s

Best ratio on every graph tested — without giving up speed.

Traversals are >90% of a GFA and collapse under iterative 2-mer grammar compression; everything else gets columnar, type-specific encoding. Both directions are linear-time and fully parallel.



SIX ANALYSES, ONE STREAMING DECODER		
GFAZ COMMAND	REPRODUCES	ACCUMULATOR
deconstruct	vg deconstruct	per-site allele table
growth	Panacus	coverage → histogram
depth	odgi depth	per-node counters
stats	odgi stats	metadata totals
pav	odgi pav	node → groups CSR
similarity	odgi similarity	node → groups CSR



GFA → VCF WITH NO GRAPH IN MEMORY

#CHROM	POS	REF	ALT	INFO	H1	H2
chr1	4	C	T	AC=1;AN=2;AF=0.5	0	1
chr1	6	ACC	A	AC=1;AN=2;AF=0.5	0	1

Phase 1 recovers snarls from topology alone (biconnected decomposition, O(V+E), decodes zero haplotypes). **Phase 2** streams each haplotype once through a boundary state machine; a reverse boundary pair normalizes inversions.

A pangenome graph does not have to be unpacked to be queried.

Outputs match the baselines: **growth** reproduces Panacus' curve exactly, **deconstruct** agrees with vg to within 0.2% of records, **pav** matrices are structurally identical to odgi's. On the 358/369 GB HPRC graphs vg and odgi cannot load the input at all. Open: whole-genome **pav** still exceeds memory there — its node-set index is not yet compressed.